

Q6(a)(i)

Emission spectral lines will be observed.

When atoms in excited states de-excite, energy is released in the form of photons. The photon energy is equal to the difference in atomic energy levels in the transition. This produces bright lines associated with the emitted photon wavelengths.

Q6(a)(ii)

Absorption spectral lines will be observed.

When electromagnetic radiation is incident on the cool gas, photons of energy that match the energy difference between an excited state and the ground state are absorbed by the atoms in the gas. The gas

atoms then re-emit the photons in all directions, leaving dark lines of absorption in the spectrum.

Q6(a)(iii)

Both spectra consist of discrete frequencies which imply that the photons can only have particular energies.

Since atoms absorb or emit photons when electrons transit between energy states, this means that the energy levels must be discrete.

Q6(b)(i)

Continuous X-rays are produced by the slowing down of electrons when they interact with the positively charged nuclei of the metal target. The loss of KE is transformed into photon energy.

Different electrons impinging upon the metal target can have different amounts of interactions with target nuclei and hence have a distribution of decelerations. This leads to a continuous range of wavelengths produced.

Q6(b)(ii)

The new sketch should have smaller overall intensity but with the same characteristic wavelengths.

However, since the energy is halved, the cut-off wavelength is doubled.

Note:

$$E' = \frac{1}{2} E$$

$$\frac{hc}{\lambda'} = \frac{1}{2} \frac{hc}{\lambda}$$

$$\therefore \lambda' = 2\lambda$$

Q7(a)